

HW #1 – Due Wed 09/25/2024 by Midnight

In Canvas: Upload your code and a brief instruction on how to run it, together with a pdf document containing the data and plots requested below. Please upload a zip file containing all the files. The name of your files should follow this rule: *family name_first name* (e.g. Bezo_Nicola_HW1.zip).

NB: Homework is individual!

Implement a **non holonomic** vehicle in Matlab as explained in class with the following specs

- max steering angle = $\pi/4$
 - max speed = 5 m/s
 - workspace dimension 200m x 200m. The global frame is at (0,0)
 - create and represent a desired shape for your robot (rectangle, star,...) but **not a triangle**
 - The robot's initial configuration is (100, 100, $\text{rand}(\theta)$)
1. Implement a **homogeneous transformation** using a Matlab function and test it by translating the robot by ($x=20$, $y=30$) and rotating by ($\pi/6$). Plot the initial configuration and the final configuration in the workspace. You can modify and adapt the provided code **(10 pt)**
 2. **Go to goal.** In this simulation pick a random goal point in the environment and drive the robot to the point as explained in class. The initial pose of the robot is as calculated in 1. The robot initial and final velocity have to be 0. Plot x and y position, and velocities of the robot with respect to time. **(30 pt)**
 3. Implement a **cruise controller** (i.e., PID as seen in class) with $v_{\text{ref}} = 3$ m/s. Make the robot go through 3 random points in the environment without stopping switching to next goal once the current goal has been visited. Assume that the velocity has the following dynamics $v(t + dt) = v(t) + u(t) - 0.01v(t)$ where u is the output of the PID and $dt=0.1$. Plot the velocity of the robot. **(30pt)**
 4. **follow a wall:** draw a line in the workspace and make the robot follow the line at a constant distance (10m) from the line, as explained in class. Plot the relative distance between robot and line over time **(30pt)**
 5. **Extra points (10 pt) Pick either: A) Move to pose:** robot starts in position (10,10,0) and needs to go to (130, 70, $\pi/4$). Plot the initial pose, trajectory, and final pose, OR **B) Follow a circle:** draw a circle of radius $r = 75\text{m}$ in the center of the workspace and make the robot follow it from any starting point of the workspace. Plot the position and velocities of the robots. Everything has to be implemented as explained in class.